KAVIN KUMAR M S

8248961288 — kavinkumarms2004@gmail.com — Erode, TN 638316
linkedin.com/in/kavinkumarms — github.com/kavinkumarms4

CAREER SUMMARY

Electronics and Communication Engineering graduate specializing in Design Verification (DV) with practical experience building layered testbenches, implementing Constrained Random Verification (CRV), and writing SystemVerilog Assertions (SVA). Backed by industry-oriented SoC training and a state-level hardware debugging award, seeking an entry-level design verification role to automate and verify complex digital architectures.

TECHNICAL SKILLS

- **HDL/HVL:** SystemVerilog, Verilog, C/C++
- **EDA Tools:** Xilinx Vivado, Verilator, Icarus Verilog, GTKWave, Synopsys VCS, Cadence Xcelium (via EDA Playground)
- **Core DV Concepts:** Constrained Random Verification, Functional Coverage, Assertions (SVA), UVM, ASIC/SoC flow
- **Protocols:** PCIe, AMBA APB, AXI, AHB, SPI, UART
- **Hardware & Architecture:** Digital Design, CMOS Logic, FPGA Architecture (Artix-7)
- **Scripting:** Python, Bash

EXPERIENCE

IC Chip Design and Verification Trainee Aug 2025 – Dec 2025
Edveon Technologies Chennai

- Developed and executed a verification plan for the Traffic Light Controller using a layered UVM-style testbench (driver, monitor, scoreboard); achieved 95% statement and branch coverage.
- Designed a UART TX/RX with configurable baud rate and FSM-based architecture; verified functionality across standard baud rates.

SoC Design Intern July 2025 – Aug 2025
CoreEL Technologies Bangalore

- Completed certified Xilinx SoC Design Flow training through an AMD-authorized training provider.
- Designed a clock frequency divider in Verilog, synthesized it, and validated hardware behavior on a PYNQ-Z2 board.

PROJECTS

PCIe Core Enumeration Verification May 2026 – Present
Key Skills: SystemVerilog, Verilator, GTKWave, Bash Scripting, PCIe Base Specification

- Constructed a SystemVerilog verification environment to validate PCIe configuration space and device enumeration.
- Automated RTL compilation, regression test execution, and log parsing via Bash scripts, reducing manual runtime by approximately 70%.
- Injected protocol constraint violations to ensure robust error handling across transaction and data link layer boundaries.

Design and Verification of a Network-on-Chip (NoC) Router Jan 2026 – Mar 2026
Key Skills: SystemVerilog, Round-Robin Arbitration, Multi-Port Switch

- Implemented and simulated a 5-port packet-switched NoC router in SystemVerilog; applied UVM architecture (agent, driver, monitor, scoreboard).
- Built a constrained-random testbench verifying packet routing, credit-based flow control, and round-robin arbiter fairness; achieved 92% functional coverage.
- Authored SVA properties to trace crossbar interconnect and verify zero packet-drop conditions under back-pressure.

EDUCATION

B.E. in Electronics and Communication Engineering 2022 – 2026
Government College of Engineering, Salem
CGPA: 8.26

TECHNICAL ACHIEVEMENTS & PORTFOLIO

- **Hackathon Winner:** Secured 2nd Place in the state-level DebugIT Hackathon conducted by Edveon.
- **Hardware Automation:** Developed a MOSFET-based automated water level controller system.
- **Structured Code Repository:** Maintain an organized GitHub portfolio documenting modular RTL designs and verification testbenches.
- **Simulation & Debugging:** Documented dynamic waveform verifications and diagnostic log using GTKWave.